

Joongmin Shin

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Research Interests

I am a Senior Researcher at the Human-Inspired AI Research Institute, Korea University, where I founded and lead **KUDoc**. Co-advised by **Prof. Jaehyung Seo** and **Prof. Heuseok Lim**, I start from real-world needs and study how advances in AI capability can become systems that people can use and rely on.

My training in parsing taught me to think structurally; my later work in human-inspired AI taught me what that structure should ultimately serve. I have since built a connected research program across multimodal and agentic AI, using structure, retrieval, hierarchical and knowledge graphs, and memory as methodological tools. My current work asks how these systems should **represent, select, update, and verify** information so that increasingly capable AI remains useful and reliable over longer interactions.

My research has yielded **15 publications**, including **6 first-author papers** at **NeurIPS, ACL, CVPR**, and **EMNLP**, one of which was selected for an **ACL 2026 oral presentation**. I also hold **5 patents** and have received **7 awards**, including **3 paper awards**.

Multimodal Parsing · Retrieval · Reasoning *over documents and video*
Agentic AI Hierarchical & Knowledge Graph · Memory · Verification

Selected Publications

* First author or co-first author

ADMIT: Support-Gated Memory-Write Admission for Document QA Agents

Joongmin Shin*, Gyuho Shim, Hyeonseok Moon, Jaehyung Seo
NeurIPS 2026 – Conference on Neural Information Processing Systems (**Main**)

PILAR: A Page-Grounded Unified Evidence Representation via an Entity-Linked Assertion Graph for Open-Domain QA Agents over Multimodal Document Corpora

Joongmin Shin*, Gyuho Shim, Jung-Hun Lee, Jaehyung Seo
EMNLP 2026 – Conference on Empirical Methods in Natural Language Processing (**Findings**)

HiKEY: Hierarchical Multimodal Retrieval for Open-Domain Document Question Answering

Joongmin Shin*, Gyuho Shim, Jeongbae Park, Jaehyung Seo, Heuseok Lim
ACL 2026 – Annual Meeting of the Association for Computational Linguistics (**Main, Oral**)

M3DocDep: Multi-modal, Multi-page, Multi-document Dependency Chunking with Large Vision-Language Models

Joongmin Shin*, Jeongbae Park, Jaehyung Seo, Heuseok Lim
CVPR 2026 – IEEE/CVF Conference on Computer Vision and Pattern Recognition (**Main**)

MultiDocFusion: Hierarchical and Multimodal Chunking Pipeline for Enhanced RAG on Long Industrial Documents

Joongmin Shin*, Chanjun Park, Jeongbae Park, Jaehyung Seo, Heuseok Lim
EMNLP 2025 – Conference on Empirical Methods in Natural Language Processing (**Main**)

Intelligent Predictive Maintenance RAG Framework for Power Plants: Enhancing QA with StyleDFS and Domain Specific Instruction Tuning

Seongtae Hong*, Joongmin Shin*, Jaehyung Seo, Taemin Lee, Jeongbae Park, Cho Man Young, Byeongho Choi, Heuseok Lim
EMNLP 2024 – Conference on Empirical Methods in Natural Language Processing (**Industry Track**)

Search-Based Generation Techniques for Improving LLM Responses: A Comparative Study of Zero-shot and RAG on GPT-3.5 and GPT-4

Joongmin Shin*, SeongReol Park, Jung-Hun Lee
KIICE 2023 – Korea Institute of Information and Communication Engineering
Award: Outstanding Paper Award

EDT5: Proposed Embedding Model of T5 Encoder-Decoder Structure

Joongmin Shin*, Jugyung Jung, Jung-Hun Lee, Sanghyun Cho, Minho Kim, Miyeon Kim, Hyuk-Chul Kwon
KSC 2022 – Korea Software Congress
Award: Outstanding Paper Award

A Dependency Parsing Model with Reinforced Head-Dependent Constraint Rules: Combining Deep Learning and Linguistic Knowledge

Joongmin Shin*, Hyuk-Chul Kwon
HCLT 2022 – Conference on Human and Cognitive Language Technology
Award: Outstanding Research Award

Research Experience

KUDoc: Korea University Document AI Research Group Seoul, Republic of Korea
FOUNDER & LEAD 2024 – Present

- **Founding & research agenda:** Built the group from the ground up as an independent line of document AI work at Korea University.
- **Project coordination:** Oversaw KUDoc-affiliated projects that produced 15 publications and multiple industry collaborations.
- **Technical direction:** Steered research across multimodal document parsing, structure-aware retrieval, grounded document QA, and agentic systems.

Human-Inspired AI Research Institute, Korea University Seoul, Republic of Korea
SENIOR RESEARCHER Jun 2023 – Present

- **Structure-grounded multimodal reasoning:** Developed retrieval and reasoning methods that preserve document structure – layout, tables, figures, and cross-page dependencies – turning noisy multi-page documents and video into grounded evidence for QA.
- **First-author work:** Led first-author papers at NeurIPS, ACL, CVPR, and EMNLP on agent memory-write admission, page-grounded evidence graphs, hierarchical retrieval, multimodal document dependency parsing, and structure-preserving chunking.
- **Hybrid reasoning systems:** Built QA pipelines combining LLMs/LVLMs, knowledge graphs, symbolic rules, and evidence filtering for accurate, interpretable answers.
- **Applied validation:** Validated multimodal RAG in insurance, manufacturing, and healthcare, including an insurance document QA MVP with Samsung Fire & Marine.

AI Lab, Pusan National University Busan, Republic of Korea
GRADUATE RESEARCHER Sep 2020 – Feb 2023

- **Korean dependency parsing:** Developed neural-symbolic dependency parsing methods that integrate morphological and syntactic information for robust Korean language understanding.
- **Table and document QA:** Designed pipelines that convert semi-structured tables and documents into structured representations for question answering and interpretable reasoning.
- **Applied collaborations:** Contributed to Korean TTS prosody prediction with KT AI2XL and additional projects in finance QA, sentiment-based marketing, and recommendation systems.

Education

Pusan National University Busan, Republic of Korea
M.S. IN ARTIFICIAL INTELLIGENCE Mar 2021 – Feb 2023

- **Advisor:** Prof. Hyuk-Chul Kwon | AI Research Lab
- **Thesis:** *A Neural-Symbolic Model for Overcoming Deep Learning Limitations in Korean Dependency Parsing*
- **Selected outcomes:** Developed neural-symbolic Korean dependency parsing and table/document QA methods; received Outstanding Paper and Student-led Research Project awards.

Pusan National University Busan, Republic of Korea
B.S. IN COMPUTER SCIENCE AND ENGINEERING Mar 2016 – Feb 2021

- **Undergraduate Research:** Prof. Jeongu Kim’s group and Prof. Hyuk-Chul Kwon’s AI Research Lab
- **Selected outcomes:** Worked on knowledge-graph-based Korean NLP, Korean grammar correction, Korean TTS, and transformer-based Korean language models; received a merit-based Milestone Scholarship.

Selected Honors & Awards

2024	Deputy Prime Minister and Minister of Education Award , Competition for Best Practices in Academia–Industry Cooperation	National Research Foundation (NRF)
2023	Outstanding Paper Award , Korea Institute of Information and Communication Engineering (KIICE)	KIICE
2023	Grand Prize , Samsung Fire & Marine Insurance LLM-based MVP Model Idea Competition	Samsung
2023	Excellence Award (Technology Track) , KU-Grant Program	Korea University
2022	Outstanding Paper Award , Korea Software Congress (KSC)	KSC
2022	Outstanding Research Award , Student-led Research Project, Graduate School of AI	Pusan National University
2020	Milestone Scholarship (Grade A) , SW-centered University Milestone Scholarship (merit-based)	Pusan National University

Patents

2026	Hierarchical Multimodal Retrieval for Open-Domain Document Question Answering	Korea, Pending
2026	Multi-modal, Multi-page, Multi-document Dependency Chunking with Large Vision-Language Models	Korea, Pending
2025	Device and Method for Document Chunking, and Question-Answering Device and Method Using the Same	Korea · DP-2025-0093, Pending
Feb 2023	Rule Filtering Techniques and Methods for Knowledge Inference Systems Based on Deep Learning	Korea · P2022-0340-KR00, Registered
Nov 2022	AI-Based Adaptive Problem Recommendation System and Method for Learner-Level Assessment	Korea · 10-2022-0068075, Registered

Technical Skills

Modeling	LLM/LVLM fine-tuning, hierarchical retrieval, knowledge graphs, hybrid neural-symbolic reasoning, rule-based evidence filtering
Document AI	Multimodal parsing, hierarchical chunking, layout-aware processing, table/document QA
Programming	Python, SQL, C/C++, Java
ML & LLM Frameworks	PyTorch, HuggingFace Transformers, PEFT/LoRA, Accelerate, OpenAI/Anthropic API
Retrieval, RAG & Evaluation	FAISS, Elasticsearch, Pinecone, LangChain, LlamaIndex, lm-eval-harness
Document Processing	PyMuPDF, pdfplumber, Tesseract, PaddleOCR, LayoutLMv3, Donut
Languages	Korean (native), English (professional working proficiency)

Teaching & Mentoring

KUDoc, Korea University	Seoul, Republic of Korea
RESEARCH MENTOR	2024 – Present
- Mentor 3 M.S. students and 2 Ph.D. students on document AI, multimodal RAG, and structured evidence reasoning. - Co-authored 15 publications and 4 manuscripts currently under review at top-tier venues.	
Pusan National University	Busan, Republic of Korea
TEACHING ASSISTANT & RESEARCH MENTOR	2019 – 2022
Capstone research mentoring (2021–2022): Mentored undergraduate capstone teams on Korean dependency parsing and Table QA; one team achieved SOTA on the KLUE-DP benchmark. PNU Software Hackathon (2021): As TA for Prof. Hyuk-Chul Kwon’s Software Design course, mentored 100+ students across 30+ projects; seven student teams won awards including the Grand Prize. Course teaching assistant: Software Design (Android, Java) and Web Application Programming under Prof. Hyuk-Chul Kwon; Programming Principles (C) under Prof. Eunmi Jeong.	

Selected Research Projects

MULTIMODAL REASONING & DOCUMENT AI

Video Curation System – Multimodal Video QA

LEAD RESEARCHER · KOREA UNIVERSITY	Jul 2025 – Present
Video QA and RAG system for structured curation and grounded retrieval over long multimodal content.	

DocGraph Copilot – Structure-Aware Multimodal RAG

LEAD RESEARCHER · KOREA UNIVERSITY	Jan 2025 – Present
Structure-aware multimodal chunking pipeline for grounded retrieval and long-document QA; produced first-author papers at CVPR 2026 and EMNLP 2025.	

Science-Domain Korean LLM & RAG Pipeline

LEAD RESEARCHER · KOREA UNIVERSITY	Feb 2024 – Dec 2024
Korean LLM and RAG pipeline specialized for the science domain; led data preprocessing and chunking methodology development; produced an EMNLP 2024 Industry Track publication.	

Synerpeace Pilot – Insurance Document Assistant

LEAD RESEARCHER · KOREA UNIVERSITY · SAMSUNG FIRE & MARINE INSURANCE	Jul 2023 – Jan 2024
Insurance document QA pilot combining LLM-based retrieval with production-oriented application development.	

Synapse – Multilingual Medical Consultation System

LEAD RESEARCHER · KOREA UNIVERSITY · KU-GRANT PROGRAM	Jun 2023 – Present
Multilingual LLM-based medical consultation chatbot; selected for the KU-Grant research commercialization program.	

FOUNDATION MODELS AND ADAPTATION

National Representative K-AI Research Team

RESEARCHER · KOREA UNIVERSITY · NC AI CONSORTIUM Aug 2025 – Jan 2026
Multilingual foundation model development as a national-level representative researcher, spanning data pipelines, post-training, and evaluation.

Ko-Gemma – Korean-Adapted Gemma

RESEARCHER · KOREA UNIVERSITY Feb 2024 – Mar 2024
Korean adaptation of Google’s Gemma model; contributed Korean-focused data adaptation and Korean benchmark evaluation.

KULLM – Korea University Korean LLM

RESEARCHER · KOREA UNIVERSITY Dec 2023 – Feb 2024
Korean LLM initiative developed at Korea University; contributed instruction-following tuning and benchmark evaluation.

Robot Process Automation – PLC Control-Code Generation

RESEARCHER · KOREA UNIVERSITY · HYUNDAI MOBIS Sep 2023 – Feb 2024
GPT-based assistant that generates PLC control code for smart-factory robot process automation, in collaboration with Hyundai Mobis.

Mi:deum K 1.0 – Korean Large Language Model

GRADUATE RESEARCHER · PUSAN NATIONAL UNIVERSITY · KT Dec 2021 – Dec 2022
Contributed to KT’s Korean LLM ecosystem for enterprise generative AI; led model adaptation and Korean-targeted post-training evaluation.

Exobrain WiseQA – National QA

GRADUATE RESEARCHER · PUSAN NATIONAL UNIVERSITY · ETRI Sep 2020 – Nov 2022
Table QA and neural-symbolic Korean sentence analysis for the national Exobrain project; introduced an Adapter-based architecture unifying separate text/table embedding models that reduced QA model parameter size by approximately 50%; research foundation for subsequent structure-grounded evidence work.

ADDITIONAL PROJECTS

- Korean parsing (96.28%) and TTS prosody (92.13%) systems with KT, deployed in KT GiGA Genie.
- Navigation-TTS for Navice in BMW/Mercedes-Benz.
- AI-based online judge with NextStage in PNU SW curriculum.
- AI mineral location prediction with KIGAM.

All Publications

* First author or co-first author

INTERNATIONAL TOP CONFERENCE PUBLICATIONS

ADMIT: Support-Gated Memory-Write Admission for Document QA Agents

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Intelligent Predictive Maintenance RAG Framework for Power Plants: Enhancing QA with StyleDFS and Domain Specific Instruction Tuning

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EMNLP 2024 – Conference on Empirical Methods in Natural Language Processing (Industry Track)

JOURNAL PUBLICATIONS

Multi-Paragraph Machine Reading Comprehension with Hybrid Reader over Tables and Text

Sanghyun Cho*, SeongReol Park, Hye-Lynn Kim, Jung-Hun Lee, Joongmin Shin, Hyuk-Chul Kwon
Applied Artificial Intelligence 38.1 (2024), p. 2367820. Taylor & Francis

Distance Based Korean WordNet (alias. KorLex) Embedding Model

SeongReol Park*, Joongmin Shin, Sanghyun Cho, Hyuk-Chul Kwon, Jung-Hun Lee
Applied Artificial Intelligence 38.1 (2024), p. 2398920. Taylor & Francis

DOMESTIC CONFERENCE PUBLICATIONS

Search-Based Generation Techniques for Improving LLM Responses: A Comparative Study of Zero-shot and RAG on GPT-3.5 and GPT-4

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KIICE 2023 – Korea Institute of Information and Communication Engineering
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KSC 2022 – Korea Software Congress
Award: Outstanding Paper Award

Evaluation of Generalization Performance in Korean Table Machine Reading Comprehension across Domain-Specific Datasets

Hyelin Kim*, Sanghyun Cho, Joongmin Shin, Hyuk-Chul Kwon
KSC 2022 – Korea Software Congress

A Dependency Parsing Model with Reinforced Head-Dependent Constraint Rules: Combining Deep Learning and Linguistic Knowledge

Joongmin Shin*, Hyuk-Chul Kwon
HCLT 2022 – Conference on Human and Cognitive Language Technology
Award: Outstanding Research Award

Rule-Augmented Neural Network-Based Dependency Parsing

Joongmin Shin*, Sanghyun Cho, Bongwoo Nam, Hyuk-Chul Kwon
KCC 2022 – Korea Computer Congress

Korean Machine Reading Comprehension Using Continual Learning

Joongmin Shin*, Sanghyun Cho, Hyuk-Chul Kwon
HCLT 2021 – Conference on Human and Cognitive Language Technology

MASTER’S THESIS

A Neural-Symbolic Model for Overcoming Deep Learning Limitations in Korean Dependency Parsing

Joongmin Shin*, Hyuk-Chul Kwon
Master’s Thesis, Pusan National University, 2023

Selected Manuscripts Under Review

FIRST-AUTHOR MANUSCRIPTS

- **ICLR 2027.** Evaluating the practical utility of agentic memory.
- **ICLR 2027.** Citation grounding for video question answering.
- **TACL.** Evidence auditing and support-sensitive evaluation for multimodal QA.
- **IEEE TPAMI.** Error-propagation diagnostics for PDF-to-RAG pipelines.